

H_∞ feedback and feedforward controller design for active vibration isolators

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Abstract: Active vibration isolation is essential for industrial high-precision systems to suppress both floor vibrations and direct disturbance forces. This paper proposes a systematic H_∞ feedback and feedforward controller design to suppress both types of disturbances with minimum amplification of sensor noise. Regarding the latter, parametric disturbance and noise models are used in the design. The resulting H_∞ controller is shown to have disturbance rejection properties similar to a manually tuned controller but with less noise amplification and actuator drift.

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1. INTRODUCTION

Vibration isolation systems are widely used in high-precision machines to isolate payloads from external disturbances (Iwaya et al. (2011); Heertjes et al. (2013)). There are two types of disturbances the system has to reject: (a) floor vibrations which enter the payload via the suspension system, and (b) disturbance forces acting directly on the isolated payload (e.g. acoustic waves). In passive systems, the disturbance rejection properties strongly depend on the suspension stiffness (Beard et al. (1994)). Systems with a low stiffness (soft mounts) generally achieve a good suppression of floor vibrations but poor rejection of direct disturbances. Systems with a high stiffness (hard mounts) generally achieve a good rejection of direct disturbances but poor rejection of floor vibrations.

Feedback control can be used to reduce the sensitivity for both types of disturbances, see e.g. Preumont (2011). However, high-gain feedback can lead to undesired amplification of sensor noise. To minimize noise amplification, mixed-sensor feedback control strategies are developed that use different types of sensors dedicated to different frequency ranges. In (Zuo and Slotine (2005); Ding et al. (2010)), non-linear sliding mode control strategies are proposed that improve performance by mixing relative position feedback and absolute velocity (or acceleration) feedback. For hard mounts, an example of mixed-sensor feedback control is presented in Tjepkema et al. (2011). Disturbance feedforward control is a promising alternative for feedback control to improve rejection of floor vibrations, see e.g. Fraanje et al. (2007); Landau et al. (2011); Beijen et al. (2014); Collette and Matichard (2015). Using feedforward control, the disturbance rather than the resulting payload vibration is measured. This generally improves the signal-to-noise ratio as compared to feedback control (Van de Ridder et al. (2015)), because the measurements are not (mechanically) low-pass filtered by the passive system.

In Beijen et al. (2016), we presented a practical tuning guide for mixed feedback and feedforward control; feedback control to reject direct disturbances and feedforward control to reject floor vibrations. However, the amplification of sensor noise is not explicitly taken into account in the controller design, and therefore likely to express itself in an undesired manner. H_∞ controller design can help to find a controller that gives a well-designed tradeoff between disturbance rejection, noise amplification, and robustness for plant uncertainty. Amongst others, Chida et al. (2008); Oomen et al. (2014) designed H_∞ feedback controllers which improve disturbance rejection and appear robust to plant variation, but do not take into account noise amplification and rejection of floor vibrations. Zhang et al. (2005) presents an H_∞ control design method that takes into account disturbances, noise, and plant uncertainty to track a modeled skyhook target. However, no systematic method to design the weighting functions for explicit shaping of disturbance rejection and noise amplification properties is given.

In this paper, an H_∞ controller design method is presented for mixed feedback and feedforward control with application to industrial vibration isolators. The main contribution involves a systematic approach toward minimizing payload vibrations and actuator drift by appropriate design of the weighting functions. The design is based on identified parametric models for the plant, floor vibrations, and sensor noises. The system and the problem description are presented in Section 2. The controller design and closed-loop results are discussed in Section 3 and Section 4, respectively. The main conclusions are given in Section 5.

2. ACTIVE VIBRATION ISOLATION SYSTEM

The system under investigation is shown in Fig. 2 and consists of the following parts: (a) a 6 degree-of-freedom movable metrology frame (MF), (b) a base frame (BF) that is supported by the floor, and (c) three isolation modules (IM) that are installed between the BF and the MF. Each IM contains a pneumatically controlled air mount that provides gravity compensation, damping, and a soft

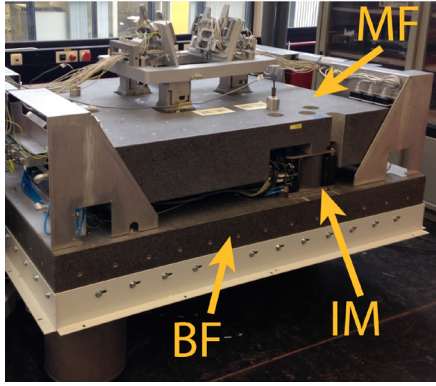


Fig. 1. Experimental setup, with a metrology frame (MF), base frame (BF), and an isolation module (IM).

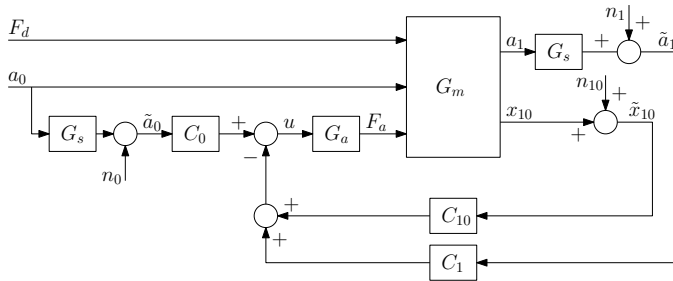


Fig. 2. Block diagram of a vibration isolation system in feedback plus feedforward configuration. Note that $a_0 = \ddot{x}_0$ and $a_1 = \ddot{x}_1$.

suspension stiffness. To enable active vibration isolation, each IM is equipped with Lorentz motors, and proximity sensors measuring the relative position between the BF and MF. Furthermore, the BF and MF are equipped with accelerometers to measure their absolute accelerations.

An overview of the closed-loop system is given in the block diagram of Fig. 2. The system model consists of the mechanics G_m , actuator dynamics G_a , and sensor dynamics G_s . Two external disturbances are considered, i.e. BF vibrations a_0 and direct disturbance forces F_d . The closed-loop system has three controllers: C_0 provides disturbance feedforward control based on a_0 , C_1 provides feedback control based on the absolute MF acceleration a_1 , and C_{10} provides feedback control based on the relative position x_{10} of the MF with respect to the BF. Note that a_0 , a_1 , and x_{10} are measured as $\tilde{a}_0$, $\tilde{a}_1$, $\tilde{x}_{10}$ with additive sensor noise represented by n_0 , n_1 , and n_{10} , respectively.

2.1 Problem description

Reconsider the signal a_1 in Fig. 2. It is assumed that a_1 can be written in the Laplace domain as

$$A_1(s) = \mathcal{T}(s)A_0(s) + s^2\mathcal{C}(s)F_d(s) + \mathcal{S}_0(s)N_0(s) + \mathcal{S}_1(s)N_1(s) + \mathcal{S}_{10}(s)N_{10}(s), \quad (1)$$

with closed-loop sensitivity functions $\mathcal{T}$ (transmissibility), $\mathcal{C}$ (compliance), and $\mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_{10}$ (sensor noise sensitivity functions). The goal is to design the controllers C_0, C_1, C_{10} such that a norm on the weighted closed-loop sensitivity functions is minimized. By doing so, a balanced tradeoff is found between rejection of the physical disturbances a_0, F_d on the one hand, and minimization of noise amplification on the other hand. Note that amplification of low-frequency noise not only leads to increased A_1 , see (1), but also leads to drift of the actuator signal which is undesired because the Lorentz actuator forces are limited to ± 30 N.

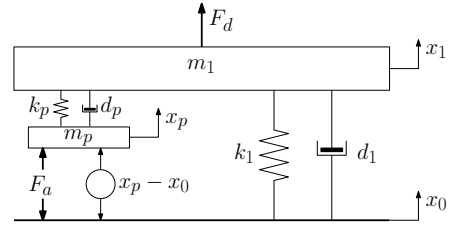


Fig. 3. Simplified model. x_0 and x_1 represent BF and MF motion, respectively. The IM has stiffness k_1 and damping d_1 . The actuator force F_a and proximity sensor are attached to a separate plate with mass m_p connected to the MF via stiffness k_p and damper d_p .

2.2 System model

This paper only considers vibration isolation for the vertical axis, for which the mechanical model is shown in Fig. 3. In this model, the BF is considered as ideal displacement source having floor displacement x_0 , and the MF is considered as a rigid-body with mass m_1 and displacement x_1 . Note that $\ddot{x}_0 = a_0$ and $\ddot{x}_1 = a_1$. The damping and stiffness of the IM are denoted by d_1 and k_1 , respectively. A direct disturbance force acting on the MF is represented by F_d . The actuator force F_a and proximity sensor measuring $x_p - x_0$ are attached to a separate plate with mass m_p that is connected to the MF via spring k_p and damper d_p . The equations of motion for the model are given by

$$\begin{bmatrix} m_p & 0 \\ 0 & m_1 \end{bmatrix} \begin{bmatrix} \ddot{x}_p \\ \ddot{x}_1 \end{bmatrix} + \begin{bmatrix} d_p & -d_p \\ -d_p & d_p + d_1 \end{bmatrix} \begin{bmatrix} \dot{x}_p \\ \dot{x}_1 \end{bmatrix} + \begin{bmatrix} k_p & -k_p \\ -k_p & k_p + k_1 \end{bmatrix} \begin{bmatrix} x_p \\ x_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} F_a \\ F_d \end{bmatrix} + \begin{bmatrix} 0 \\ d_1 \end{bmatrix} \dot{x}_0 + \begin{bmatrix} 0 \\ k_1 \end{bmatrix} x_0. \quad (2)$$

From these equations of motion, the transfer function block G_m can straightforwardly be derived.

The actuator dynamics G_a are described by a low-pass filter to account for the induction pole caused by voltage steering of the Lorentz actuators. The transfer function from control signal u to Lorentz force F_a is modeled as

$$G_a(s) = \frac{F_a(s)}{U(s)} = \frac{\omega_q}{s + \omega_q}. \quad (3)$$

Sensor dynamics G_s are only considered for the accelerometers, which contain a band-pass filter from $\omega_1 = 2\pi 0.1$ rad/s to $\omega_2 = 2\pi 450$ rad/s to suppress sensor noise at very low and high frequencies:

$$G_s(s) = \left(\frac{s^2}{s^2 + 1.4\omega_1 s + \omega_1^2} \right) \left(\frac{\omega_2^2}{s^2 + 1.4\omega_2 s + \omega_2^2} \right).$$

Numerical values for the model parameters $m_1 = 1100$ kg, $d_1 = 1100$ Ns/m, $k_1 = 140 \cdot 10^3$ N/m, $m_p = 12.2$ kg, $d_p = 1000$ Ns/m, $k_p = 6.5 \cdot 10^6$ N/m, $\omega_q = 2\pi 80$ rad/s are obtained from fitting the model on measured frequency response function (FRF) data, see Fig. 4. This figure shows that the model is a reasonable approximation of the measured FRF up to the resonance at 115 Hz. Note that the suspension frequency is at 1.8 Hz which is representative for soft-mount systems.

2.3 Disturbance models

Models for the incoming disturbances a_0, n_0, n_1, n_{10} are based on power spectral density (PSD) measurements obtained from the setup. Fig. 5 shows the measured PSDs and the fitted parametric noise models which are given by

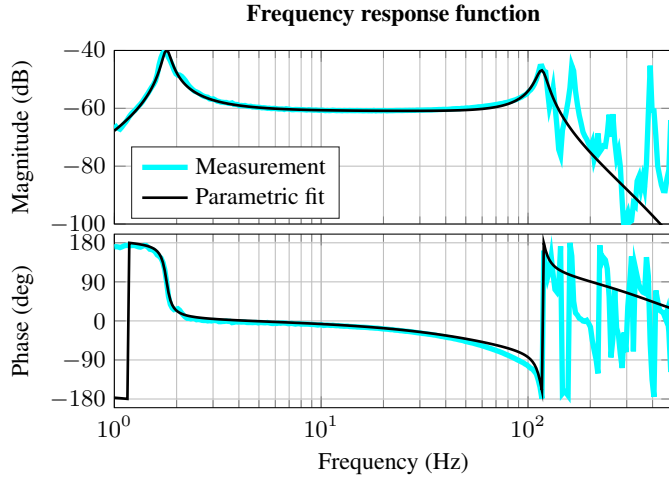


Fig. 4. Measured FRF and parametric fit from actuator signal u to MF acceleration $\tilde{a}_1$ in vertical direction.

$$\begin{aligned}
 P_{a_0 a_0}(\omega) &= 4 \cdot 10^{-9} \left| \frac{j\omega}{(j\omega + 2\pi 4)} \cdot \frac{2\pi 100}{(j\omega + 2\pi 100)} \right|^4, \\
 P_{n_0 n_0}(\omega) &= P_{n_1 n_1}(\omega) = 4 \cdot 10^{-14} \left| \frac{(j\omega + 2\pi 3)}{\omega} \right|^2, \\
 P_{n_{10} n_{10}}(\omega) &= \frac{1 \cdot 10^{-14}}{\omega}.
 \end{aligned} \quad (4)$$

3. CONTROLLER DESIGN

In Beijen et al. (2016), feedback and feedforward controllers are designed from physical arguments. The resulting physics-driven controller is summarized in Section 3.1. This controller will be used as a reference controller to design an H_∞ controller as proposed in Section 3.2.

3.1 Physics-driven controller design

Relative position feedback is used to add virtual stiffness,

$$C_{10}(s) = K_s. \quad (5)$$

Proportional and integral acceleration feedback are used to add virtual mass and skyhook damping, respectively,

$$C_1(s) = K_a + \frac{K_v}{s}, \quad (6)$$

with gain K_a representing virtual mass, and gain K_v representing virtual skyhook damping. The idea is that virtual stiffness is used to reduce $\mathcal{C}$, and virtual mass is used to prevent that $\mathcal{T}$ increases due to virtual stiffness. Skyhook damping is used to damp the suspension mode at 1.8 Hz. To obtain robust stability with respect to non-modeled dynamics and sensor noise, the controllers are augmented with loop shaping filters:

$$\begin{aligned}
 C_{10,r}(s) &= C_{10}(s)H_{hp}(s)H_{lp}(s)H_{lead}(s), \\
 C_{1,r}(s) &= C_1(s)H_{hp}(s)H_{lp}(s)H_{lead}(s).
 \end{aligned} \quad (7)$$

The motivation for these filters is as follows. To prevent amplification of low-frequency sensor noise in acceleration feedback control, a high-pass filter at $\omega_{hp} = 2\pi 1.8$ rad/s will be used:

$$H_{hp}(s) = \frac{s}{s + \omega_{hp}}. \quad (8)$$

To limit the controller bandwidth at high frequencies, a low-pass filter at $\omega_{lp} = 2\pi 1.8$ rad/s is added:

$$H_{lp}(s) = \frac{\omega_{lp}^2}{s^2 + 2\zeta_{lp}\omega_{lp}s + \omega_{lp}^2}. \quad (9)$$

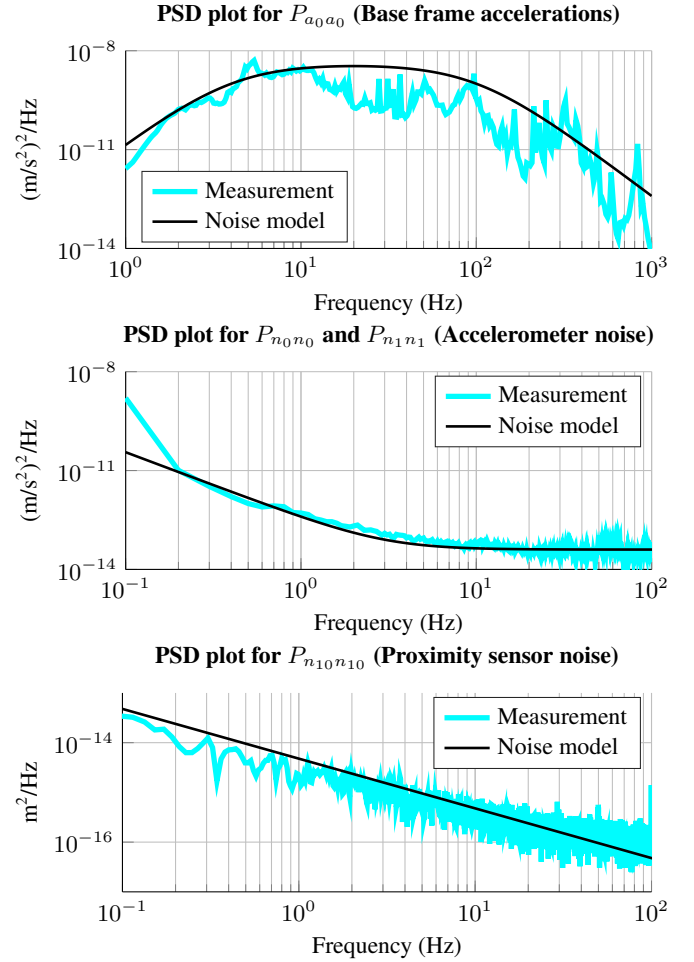


Fig. 5. Power spectral density (PSD) plots for the measured and modeled input disturbances, see (4).

To improve the phase margin at the open-loop cross-over frequency (≈ 30 Hz), a lead filter will be used:

$$H_{lead}(s) = 5 \frac{s + 80}{s + 400}. \quad (10)$$

The feedforward controller is given by

$$C_{FF,n}(s) = -H_{(\alpha,n)}^2(s)(ds + k + C_1(s)), \quad (11)$$

with

$$H_{(\alpha,n)}(s) = \frac{1 - L_{(\alpha,n)}(s)}{s}, \quad (12)$$

describing an n^{th} -order weak integrator, and

$$L_{(\alpha,n)}(s) = \left(\frac{\alpha}{s + \alpha} \right)^n, \quad (13)$$

representing an n^{th} -order low-pass filter with cut-off frequency α . The idea behind the controller in (11) is that it improves vibration isolation performance by providing an additional roll-off of order n in $\mathcal{T}$ (Beijen et al. (2016)). In this paper, the values $\alpha = 2\pi$ rad/s and $n = 2$ are used.

3.2 H_∞ controller design

The controller from Section 3.1 aims at desired performance in terms of $\mathcal{T}$ and $\mathcal{C}$. The noise sensitivity functions $\mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_{10}$, however, are not explicitly shaped. In order to minimize $\mathcal{T}$ and $\mathcal{C}$ while keeping desired properties for noise amplification, H_∞ control design will be used.

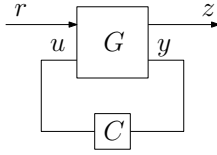


Fig. 6. General plant for H_∞ controller synthesis.

To this end, the control problem must be cast into a general control problem formulation, see e.g. Skogestad and Postlethwaite (2007). The generalized plant, whose structure is shown in Fig. 6, is described by

$$\begin{bmatrix} z \\ y \end{bmatrix} = G \begin{bmatrix} r \\ u \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} r \\ u \end{bmatrix}. \quad (14)$$

To ease notation, we dropped the explicit reference to the Laplace domain. The disturbance vector is defined as $r = [a_0, F_d, n_0, n_1, n_{10}]^T$ in order to shape $\mathcal{T}$, $\mathcal{C}$, and $\mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_{10}$. The objective vector is defined as $z = [a_1, u]^T$ to minimize MF vibrations and actuator drift, as explained in Section 2.1. The measured outputs are $y = [\tilde{a}_0, \tilde{a}_1, \tilde{x}_{10}]^T$. An expression for G can be derived from the block diagram and the models in Section 2.2.

The standard H_∞ control problem is to find a stabilizing controller C that minimizes

$$\|W F_l(G, C) V\|_\infty = \max_w \bar{\sigma}(\mathcal{W}(j\omega) F_l(G, C)(j\omega) \mathcal{V}(j\omega)), \quad (15)$$

where the closed-loop transfer function from w to z is given by the linear fractional transformation

$$\begin{aligned} z &= F_l(G, C) \cdot w \\ &= (G_{11} + G_{12}C(I - G_{22}C)^{-1}G_{21}) \cdot w. \end{aligned} \quad (16)$$

In addition, $\mathcal{W} = \text{diag}(\mathcal{W}_1, \mathcal{W}_2)$, $\mathcal{V} = \text{diag}(\mathcal{V}_1, \dots, \mathcal{V}_5)$ in (15) represent weighting functions on z and r , respectively, to shape the desired closed-loop performance of the vibration isolator. The design of these weights is discussed next.

Remark 1. In this paper, controller design is only performed for the nominal model G . However, it follows from Fig. 4 that the experimental setup has unmodeled dynamics beyond 115 Hz. Therefore, closed-loop stability will be checked afterwards via Nyquist plots, see Section 4.

3.3 Weighting functions

This section describes the design of the weights to shape the closed-loop performance functions $\mathcal{T}$, $\mathcal{C}$, and $\mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_{10}$. Relative scaling of the weights is derived from the measured PSDs in Section 2. This scaling implicitly gives the design tradeoff between maximizing vibration isolation properties and minimizing noise amplification. Bode plots of the weighting functions are found in Fig. 7.

Input weight $\mathcal{V}_1$ is used to penalize a_0 and thus prescribes the desired $\mathcal{T}$. Since $\mathcal{T}$ is affected by both feedback and feedforward control, $\mathcal{V}_1$ is composed of two parts,

$$\mathcal{V}_1(s) = c_1 \underbrace{\frac{s^2 + 1.4\omega_{z1}s + \omega_{z1}^2}{s^2 + 1.4\omega_{p1}s + \omega_{p1}^2} \frac{\omega_{p1}^2}{\omega_{z1}^2}}_{\mathcal{V}_{1,FB}(s)} \underbrace{\left(\frac{s + \alpha_1}{s + \alpha_2} \cdot \frac{\alpha_2}{\alpha_1} \right)^n}_{\mathcal{V}_{1,FF}(s)}.$$

The scaling factor $c_1 = \sqrt{4 \cdot 10^{-9}}$ is derived from the disturbance model in (4). The part $\mathcal{V}_{1,FB}$ describes the desired $\mathcal{T}$ in case only feedback control is used. In view of this, $\mathcal{V}_{1,FB}$ has unity gain below 1 Hz. By choosing $\omega_{z1} = 2\pi 1.8$ rad/s which is the passive suspension frequency, and $\omega_{p1} = 2\pi 10$ rad/s, the penalty increases with a slope of 40 dB/decade from 1.8 to 10 Hz. This corresponds

with the requirements from Beijen et al. (2016) of a decreasing $\mathcal{T}$ beyond 1.8 Hz and a damped suspension mode. The part $\mathcal{V}_{1,FF}$ represents an additional penalty to further decrease $\mathcal{T}$ using feedforward control. By choosing $\alpha_1 = \alpha = 2\pi$ rad/s, $\alpha_2 = 2\pi 5$ rad/s, and $n = 2$, an additional roll-off of order n is enforced in $\mathcal{T}$ in the mid-frequency range, which corresponds with the physics-driven feedforward control goal of (11).

Input weight $\mathcal{V}_2$ is used to penalize F_d and therefore relates to the desired $\mathcal{C}$. At low frequencies, it is desired to decrease $\mathcal{C}$ with a factor 100, see Beijen et al. (2016). To this end, $\mathcal{V}_2$ needs increased gain at low frequencies,

$$\mathcal{V}_2(s) = c_2 \frac{s^2 + 1.4\omega_{z2}s + \omega_{z2}^2}{s^2 + 1.4\omega_{p2}s + \omega_{p2}^2}, \quad (17)$$

with $\omega_{z2} = 2\pi 1$ rad/s and $\omega_{p2} = 2\pi 10$ rad/s. The scaling factor $c_2 = m_1 c_1$ is included to get the same weighting of $\mathcal{V}_1$ and $\mathcal{V}_2$, hence $\mathcal{T}$ and $\mathcal{C}$, and to compensate for the fact that at mid-frequencies $\mathcal{C}$ is given by $1/(m_1 s^2)$.

Input weights $\mathcal{V}_3$ and $\mathcal{V}_4$ are used to penalize the sensor noise on the base and payload accelerometers, respectively, and therefore relate to $\mathcal{S}_0$ and $\mathcal{S}_1$. According to Fig. 5, these sensors mainly suffer from low-frequency sensor noise which can lead to two undesired effects. First, noise amplification can deteriorate isolation performance, hence increased a_1 . Second, low-frequency noise compensation easily results in actuator drift which is undesired because the actuator forces are limited to ± 30 N. Therefore, $\mathcal{V}_3$ and $\mathcal{V}_4$ are composed of two parts to account for both effects,

$$\mathcal{V}_3(s) = \mathcal{V}_4(s) = c_3 \underbrace{\frac{s + \omega_{z1}}{s + \omega_{p1}}}_{\mathcal{V}_{3,isol}} \underbrace{\frac{s + \omega_{z2}}{s + \omega_{p2}}}_{\mathcal{V}_{3,drift}}. \quad (18)$$

The scaling factor $c_3 = \sqrt{4 \cdot 10^{-14}}$ is based on the noise model in (4). To prevent that low-frequency noise induces payload vibrations, we choose $\omega_{z1} = 2\pi 3$ rad/s corresponding to (4), and $\omega_{p1} = 0.01 \cdot 2\pi$ rad/s to create a flat slope of the weight at low frequencies which improves conditioning of the H_∞ problem. In view of minimizing drift, we further increase the penalty at low frequencies by choosing $\omega_{z2} = 2\pi$ rad/s, and $\omega_{p2} = 2\pi 0.01$ rad/s.

Input weight $\mathcal{V}_5$ is used to penalize proximity sensor noise n_{10} . The spectrum in Fig. 5 is approximated as a flat spectrum with power $10^{-14} (\text{m/s}^2)^2/\text{Hz}$, such that

$$\mathcal{V}_5(s) = c_5 = \sqrt{10^{-14}}. \quad (19)$$

Output weight $\mathcal{W}_1$ is used to penalize a_1 ,

$$\mathcal{W}_1(s) = \frac{0.5}{c_1}. \quad (20)$$

This constant is chosen such that, apart from the factor 0.5, the product $(\mathcal{W}_1 \mathcal{V}_1)^{-1}$ has unity gain for frequencies below the suspension frequency. This corresponds with a desired transmissibility of 0 dB. The factor 0.5 is included to allow a performance deterioration of a factor 2 with respect to the uncontrolled system. This value is often used in typical motion control systems, where the peak value of the sensitivity is limited to 2 (or +6 dB). Performance deterioration is inevitable when using causal feedforward and feedback controllers, see e.g. Beijen et al. (2015).

Output weight $\mathcal{W}_2$ is used to penalize the control signal u ,

$$\mathcal{W}_2(s) = c_6 \frac{s^2 + 1.4\omega_{z2}s + \omega_{z2}^2}{s^2 + 1.4\omega_{p2}s + \omega_{p2}^2} \frac{\omega_{p2}}{\omega_{z2}}. \quad (21)$$

The factor $c_6 = 1/|\mathcal{V}_2(0)|$ is chosen such that the product $\mathcal{W}_2 \mathcal{V}_2$ has unity gain at low frequencies. By doing so,

the sensitivity from F_d to u becomes unity, allowing full ability to reject direct disturbances. The filter with $\omega_{z2} = 2\pi 10$ rad/s and $\omega_{p2} = 2\pi 2000$ rad/s is used to enforce low-pass filtering in the controllers to limit the bandwidth.

4. RESULTS

Bode magnitude plots to express the vibration isolation performance in terms of the functions $\mathcal{T}, \mathcal{C}, \mathcal{S}_0, \mathcal{S}_1, \mathcal{S}_{10}$ are shown in Fig. 7. The figure also shows the desired performance specifications which are calculated from the weighting functions. Following the reasoning in Beijen et al. (2016), $\mathcal{T}$ should have sufficient roll-off beyond 1 Hz, while $\mathcal{C}$ should be improved with a factor 100 (40dB) in the mid-frequency range. It follows from Fig. 7 that both the physics-driven and H_∞ controller designs are able to meet these performance specifications at almost all frequencies. In view of noise amplification, the lower left plot shows that the H_∞ controller generally has a lower $\mathcal{S}_0$. The lower middle plot for $\mathcal{S}_1$ shows that the H_∞ controller leads to less amplification at very low frequencies, more amplification in the range 0.1–1 Hz, and similar noise amplification for higher frequencies. The lower right plot for $\mathcal{S}_{10}$ shows that the H_∞ controller clearly outperforms the physics-driven controller because the noise amplification is much lower for frequencies beyond 1 Hz.

In terms of minimizing actuator drift, it is expected from Fig. 7 that the H_∞ controller performs significantly better than the physics-driven controller, because $\mathcal{S}_0$ and $\mathcal{S}_1$ are much lower at very low frequencies. This result is confirmed by a time simulation plot over 50 seconds, see Fig. 8, where the disturbances a_0, n_0, n_1, n_{10} are modeled as stochastic signals with PSDs similar to the models in (4). The simulation results show that the residual vibration of a_1 is slightly better with the H_∞ controller; root-mean-square (RMS) values for a_1 are $1.1 \cdot 10^{-5}$ m/s² for the physics-driven controller and $1.0 \cdot 10^{-5}$ m/s² for the H_∞ controller. Regarding drift, the H_∞ controller shows a significant improvement in RMS value (0.13 V) compared to the physics-driven controller (0.93 V).

Closed-loop stability is checked from the Nyquist plots using the open-loop systems with either the physics-driven or the H_∞ controller, see Fig. 9. It is observed that for both controllers the Nyquist curve enters the unit circle with sufficient phase margin and do not encircle the point -1 . The phase margin of the H_∞ controller is better than that of the physics-driven controller. Furthermore, both Nyquist curves leave again the unit disc due to the resonances beyond 100 Hz. Although this does not lead to instability in this case, it may lead to stability problems in practice in view of plant uncertainty. Therefore, future research should focus on including this plant uncertainty in the controller design. Other aspects of ongoing research are model order reduction of the H_∞ controller, which currently has twice as much states as the physics-driven controller, and a multiple-axis implementation of the control strategy on the experimental setup.

5. CONCLUSIONS

This paper provides a systematic method to design mixed feedback and feedforward controllers for active vibration isolators. Minimization of the H_∞ norm is used to design the controllers. Design and scaling of the weighting functions are based on measured power spectral density data. Closed-loop performance of the resulting H_∞ controller is compared with a physics-driven controller. It is shown that both controllers lead to similar performance in terms

of transmissibility and compliance, but that the H_∞ controller has less amplification of accelerometer noise at low frequencies and proximity sensor noise at high frequencies. Furthermore, it is shown that there is less actuator drift when using the H_∞ controller.

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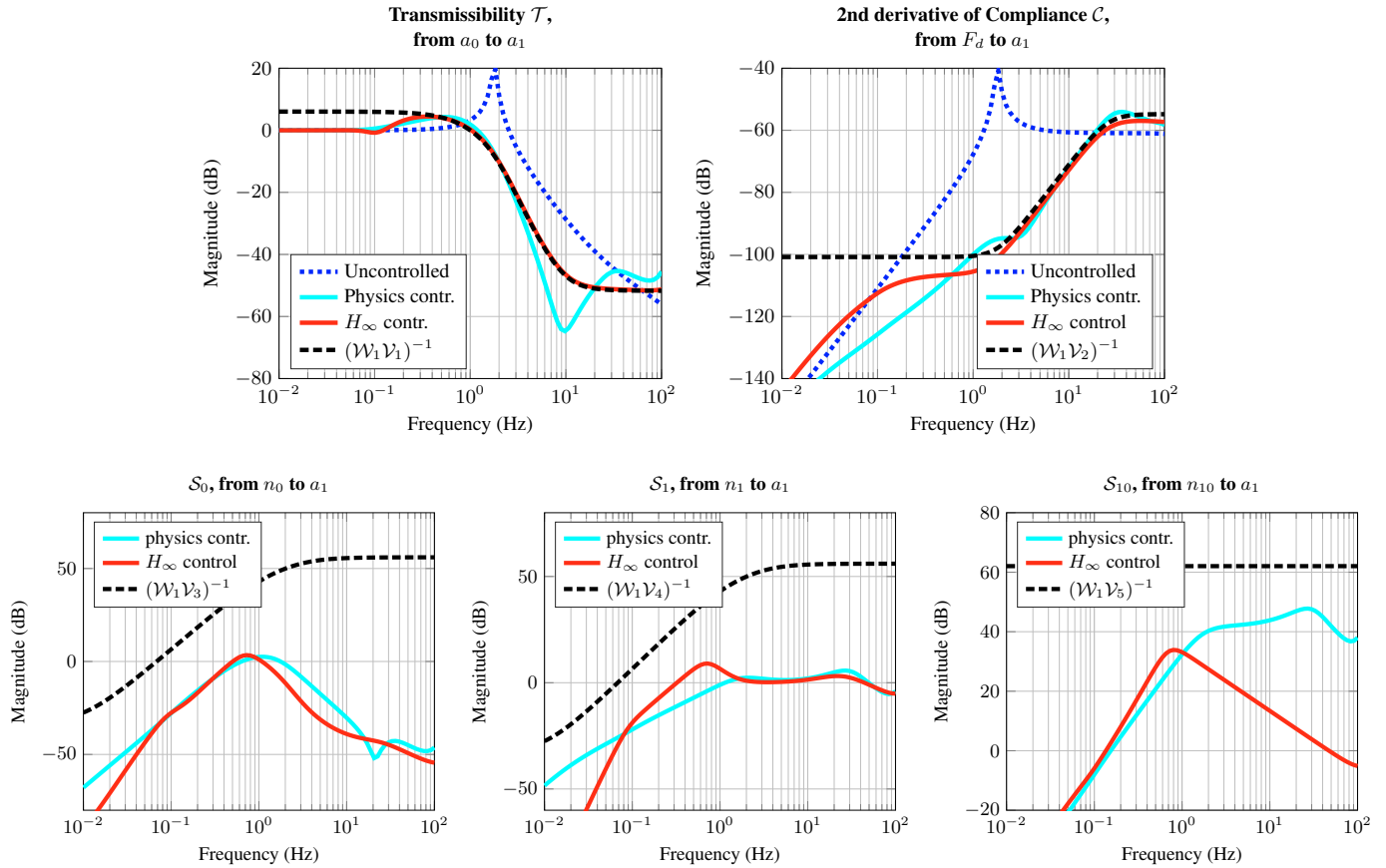


Fig. 7. Vibration isolation performance for the passive and active systems with either the physics-driven or H_∞ controller. The black dashed lines indicate performance specified by the weighting functions. In terms of $\mathcal{T}$ and $\mathcal{C}$, both the physics-driven and H_∞ controller significantly improve the performance. In terms of noise amplification, it is observed that the H_∞ controller benefits from a lower $\mathcal{S}_0$, $\mathcal{S}_1$ at low frequencies and a lower $\mathcal{S}_{10}$ at high frequencies.

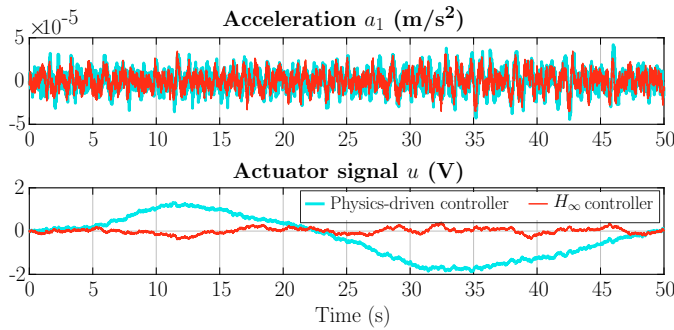


Fig. 8. Time domain simulations with both the physics-driven and H_∞ controller. Note that both controllers have similar residual payload vibration a_1 , but H_∞ control has much less actuator drift in u .

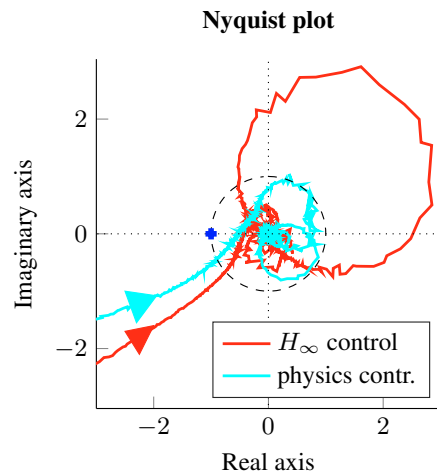


Fig. 9. Nyquist plots of the open-loop systems with either the physics-driven or H_∞ controller to show closed-loop stability.

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